Your grade: 100%

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Next item →

1. In a scenario where k-means clustering is applied to a dataset with features having very different value ranges, what preprocessing step is necessary to ensure more reliable clustering results? 1 / 1 point

- Remove outlier data points
- Normalize the feature values
- Increase the number of clusters
- Use a different clustering algorithm

Nice work

Correct. K-means clustering is sensitive to the range of feature values. Normalizing the feature values, such as using min-max scaling, ensures that each feature contributes equally to the distance calculations, leading to more reliable clustering results.

2. During a k-means clustering process, after several iterations, the centroids have stabilized and do not move significantly. What is the most likely next step in the clustering process? 1 / 1 point

- Stop the iterations
- Recalculate the distances
- Increase the number of centroids
- Decrease the number of centroids

Nice work

Once the centroids in a k-means clustering process have stabilized and show minimal or no movement between iterations, it indicates that the solution has converged. The typical next step is to stop further iterations as continuing would not significantly change the clusters.

3. How is the number of clusters determined in the k-means algorithm? 1 / 1 point

- The number of clusters must be predefined
- The number of clusters is calculated adaptively during the clustering process and usually converges toward a stable estimate at termination.
- It is based on data size

Nice work

In the K-means clustering algorithm, the number of clusters, denoted as K, must be specified in advance before the algorithm is run. This predefined number of clusters is a crucial parameter that influences the formation of clusters.

4. A data scientist is using k-means clustering to analyze customer segmentation data. After initializing three random centroids and assigning cluster memberships in the first iteration step, they notice that one centroid has no points assigned to it. What should the data scientist expect in the very next iteration regarding the unassigned centroid? 1 / 1 point

- The centroid will be removed from the analysis
- The centroid will randomly reinitialize
- The centroid will not move
- The centroid will merge with the nearest centroid

Nice work

In k-means clustering, centroids are recalculated as the mean of all points assigned to that cluster. If no points are assigned to a centroid, it has no data to recalculate its position, hence it will not move.

5. Which of the following statements about k-means clustering is *incorrect*? 1 / 1 point

- The k-means algorithm will converge toward the same globally optimal solution regardless of how the centroids are initialized.
- k-means clustering works well for simple clusters that are same size, and that have well-separated, globular shapes.
- K-means clustering does not assume a tree structure.
- K-means clustering is a partitional clustering approach, i.e. a 'hard' clustering of the data points, because each point can belong to only one cluster.

Nice work

This is false: different clusterings may result from different initializations of the centroids.